Dingwei Guo

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Summary

Ph.D. candidate in Economics with hands-on experience in

- **Experimentation & Causal Inference**: A/B testing and randomized field experiments; causal inference from observational data (Difference-in-Differences, Matching and Weighting, Synthetic Control, Instrumental Variables, Regression Discontinuity).
- **Model Improvement & Validation**: Double ML; Causal/IV Forest for heterogeneous effect estimation; Bayesian updating with hierarchical modeling for observational model calibration and validation.
- **Causal Forecasting**: Extrapolation of short-run experimental results using surrogate models; time-series forecasting.
- **Programming & Data Tools**: Proficient in SQL, Python (pandas, scikit-learn, statsmodels), and R; AWS (including Redshift) for data processing and storage.
- **End-to-End Product Analytics**: Cross-functional collaboration with stakeholders and delivery of end-to-end solutions from problem formulation to measurable business impact.

Education

- Ph.D. in Economics, University of North Carolina at Chapel Hill — *Expected May 2027; early possible in Dec 2026*
- M.S. in Economics, University of Wisconsin-Madison — *May 2022*
- B.A. in Economics, Xi'an Jiaotong University — *May 2021*

Industry & Academic Experience

- **Amazon | Prime Video Finance & Advertising | Economist Intern** — *May 2026 - Aug 2026*
  - Designed A/B experiments and Double ML-enhanced Regression Discontinuity models to estimate the causal impacts of Prime Video in-stream ads on streaming engagement with AWS tools, SQL, and Python; calibrated observational estimates through Bayesian updating.
  - Estimated heterogeneous treatment effects across content categories and customer segments using IV Forest to optimize ad allocation and targeting.
  - Built an automated pipeline and dashboard to track campaign performance and long-term advertising impact.
  - Developed a surrogate-based causal forecasting model integrating experimental and observational data to predict long-term financial outcomes beyond the experimental horizon.
  - Delivered actionable recommendations to senior product and engineering leaders, reducing ineffective ad frequency and improving ad guardrails.
- **Quasi-Experimental Policy Evaluation for Labor Market Outcomes | Ph.D. Researcher**
  - Designed five quasi-experimental policy evaluations to estimate the short-term and long-term labor market impacts of public finance and social assistance programs.
  - Built multi-source policy evaluation datasets by linking household-level panel data with macro indicators and policy information extracted from government documents using web scraping and text mining.
  - Applied propensity score matching, difference-in-differences, and instrumental variables to estimate causal policy impacts.
  - Presented findings to local government agencies and delivered evidence-based recommendations for policy targeting.
  - Publications (forthcoming) in *Journal of Development Effectiveness*, *Annals of Public and Cooperative Economics*, *Applied Economics*.
- **Experimental and Structural Modeling of Consumer Choice | Ph.D. Researcher**
  - Implemented randomized valuation surveys and choice experiments to evaluate consumer responses to energy-saving programs in electricity markets.
  - Developed structural utility models and estimated behavioral parameters from experimental data using maximum likelihood and moment-based methods.
  - Publications (forthcoming) in *Frontiers in Energy Research*, *Resource and Energy Economics*, *Environment and Development Economics*.

Awards

- 2026 Best Research Awards in Applied Micro — *UNC Chapel Hill*
- Graduate Fellowship — *UNC Chapel Hill*